Surface Area of Right Prisms and Right Cylinders

Tuesday, May 19, 2026 10:00 AM

Consider this: how do manufacturer determine the material needed to produce …





<u>Surface area</u>: total area of a 3D object's surfaces

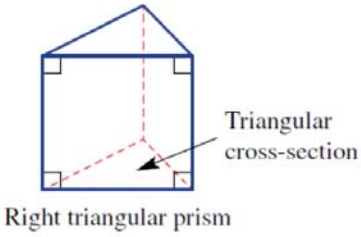


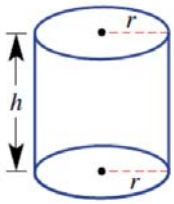
Your boss announces:
Our company is introducing Soda Mini! It's half as much soda and equally expensive
Also, it'll require half as much tin to make the can

Q: what's wrong with this statement?

<u>Right prisms</u>: a 3D object with uniform cross sections, 2 identical ends and the remaining surfaces are rectangles.

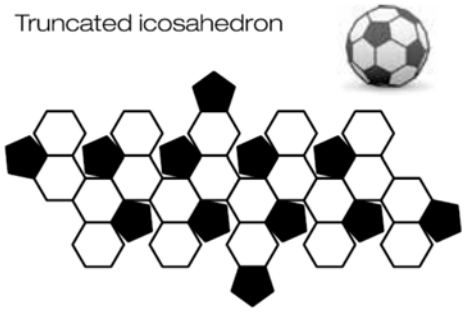
<u>Cylinders</u>: a 3D object with uniform cross sections which are disks





Because 3D shape can only be represented via projection on 2D paper, it does not give an accurate picture of the area of each face …

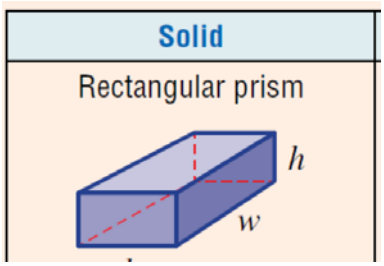
<u>Net of a 3D shape</u>: a 2D pattern consisting of polygons joining at common edges, if you cut out of the paper, fold by the joining edges and glue along some other edges, it recovers the 3D shape we are interested in

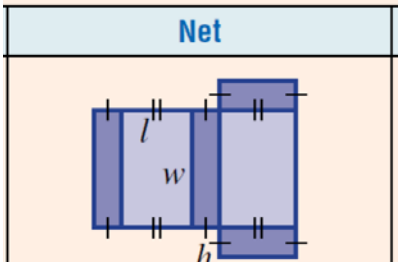


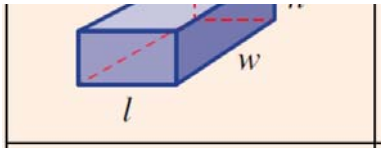
What is it good for?

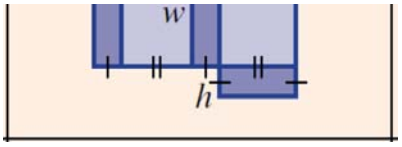
We can use net to *keep track of faces* of a 3D shape when calculating surface area!

Surface area of a right rectangular prism (cuboid)





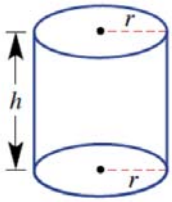


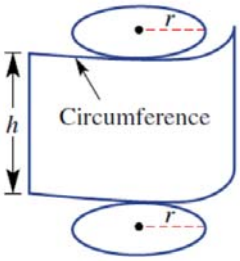


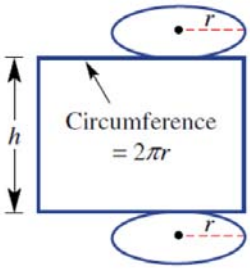
$A = 2 \times \text{top} + 2 \times \text{side} + 2 \times \text{end}$

$A = 2lw + 2wh + 2lh$

Surface area of cylinder







- Surface area of a **cylinder** = 2 circles + 1 rectangle

$$= 2 \times \pi r^2 + 2\pi r \times h$$

$$\therefore A = 2\pi r^2 + 2\pi rh$$

2 circular ends — curved area

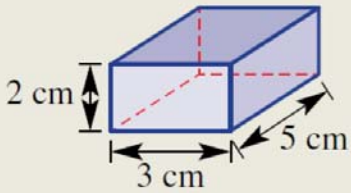
- $A = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$

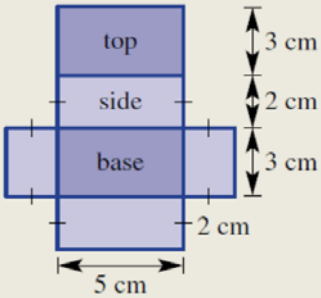
Guided exercises

#1

Find the surface area of this right rectangular prism

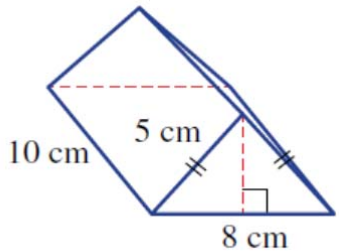
a





#2

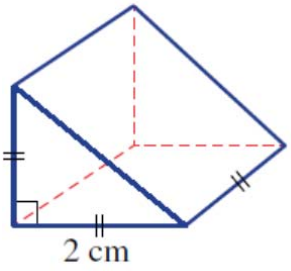
c

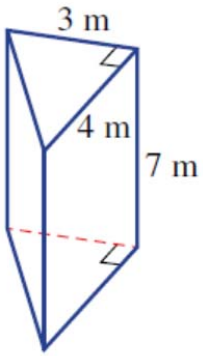


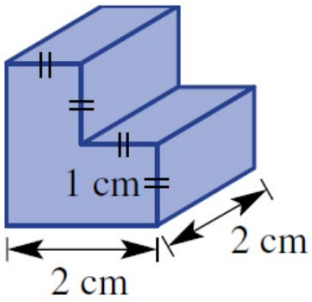
1. Draw the net
2. Set up surface area expression
3. Compute surface area

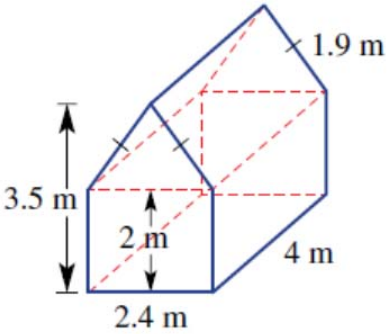
Note: clearly label each step and write words describing what we are doing in each step

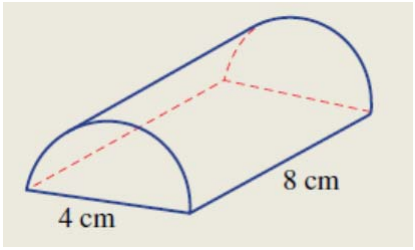
Now you do

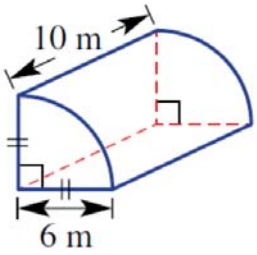

2 cm


3 m
4 m
7 m


1 cm
2 cm
2 cm


1.9 m
3.5 m
2 m
4 m
2.4 m


8 cm
4 cm


10 m
6 m